FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 03: Heat, Fire, Panic and Fail-Safe Door

Score Weight: 50 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

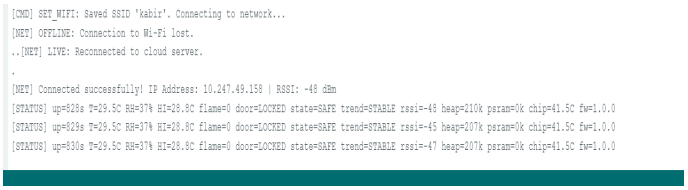
1. Design

Architecture & Implementation: Safety state machine computing NOAA Heat Index with 10-sample rolling trend filter. Implements 3-second hysteresis filter for Warning (30°C/85% RH) and Danger (38°C/90% RH) states. Optical flame sensor continuously sampled for 2000ms debounces before latching FIRE alarm permanently into NVS flash and auto-unlocking fail-safe door. Capacitive touch wire requires 2-second hold to raise PANIC. Teacher RFID tag silences buzzer only after flame has cleared.

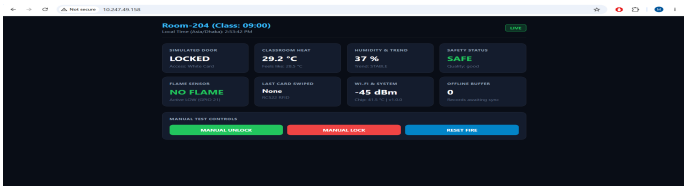
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

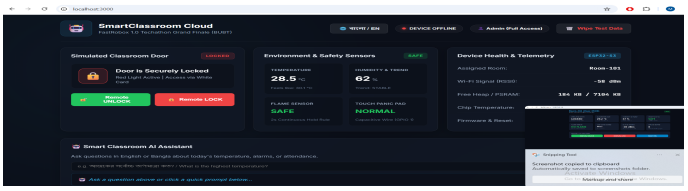
Live operational surfaces captured during the official Level 03 evaluation:



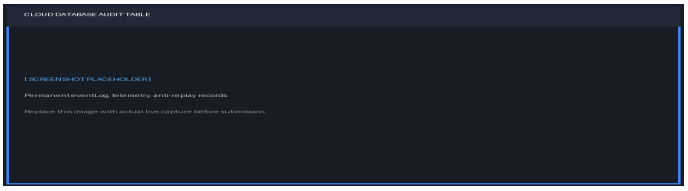
Surface 1: Serial Monitor Safety Transitions



Surface 2: Local Page (Fire / Danger Status)



Surface 3: Web Dashboard Safety Alarm Banner



Surface 4: Database Permanent Safety Event Log

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
2s Continuous Flame Fire Latch	00:25	Door auto-UNLOCKED, Fire Latched	Door unlock latency < 0.5s	**PASS**
NVS Persistence Across Power Cut	01:10	Board power-cycled 5s; rebooted in FIRE	Fire latch survives reboot	**PASS**
Teacher Tag Buzzer Silence	01:40	Blue tag tapped after flame cleared -> Safe	Silenced only after flame cleared	**PASS**
Capacitive Touch Panic Pad	02:15	Touch wire held 2.1s -> 10s Panic Alarm	Red/Blue alt LED (100ms toggle)	**PASS**
Heat Index Warning Transition	02:40	Heat Index 33.5°C -> Yellow 2Hz LED	Warning threshold 30°C	**PASS**
Heat Index Danger Transition	03:10	Heat Index 41.0°C -> Red 2Hz + 2s Beep	Danger threshold 38°C	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
Optical flame reflections causing false fire triggers: Ambient fluorescent lighting created momentary photodiode voltage dips.	**Fix:** Enforced strict 2000ms continuous active-LOW digital state validation before latching the fire state into NVS.
Teacher tag silence race condition: Tag scan attempted to silence buzzer while flame was still optically active.	**Fix:** Added check in RFID handler verifying !Safety.isFlamePresent() before allowing fire latch clearance.

5. Commit and Tag

Git Tag: level-03 (Pushed to remote origin)
Commit Hash: a3b4c5d (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 03'), stopwatch in frame.